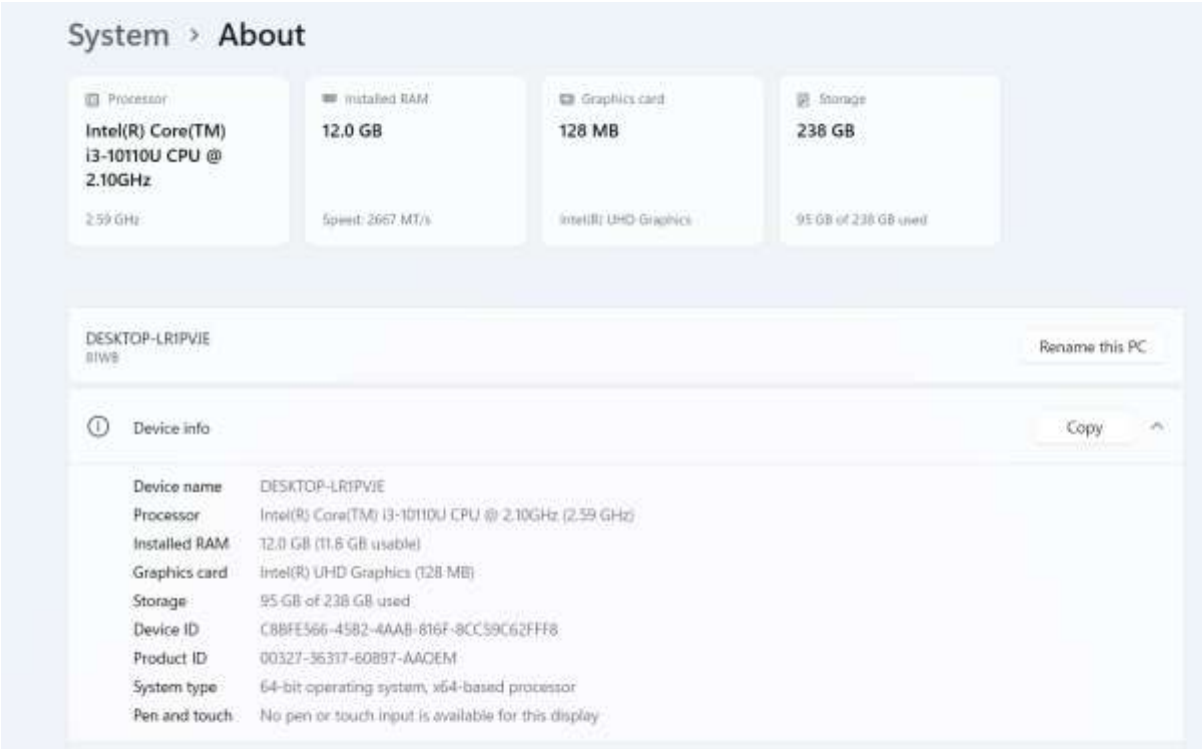
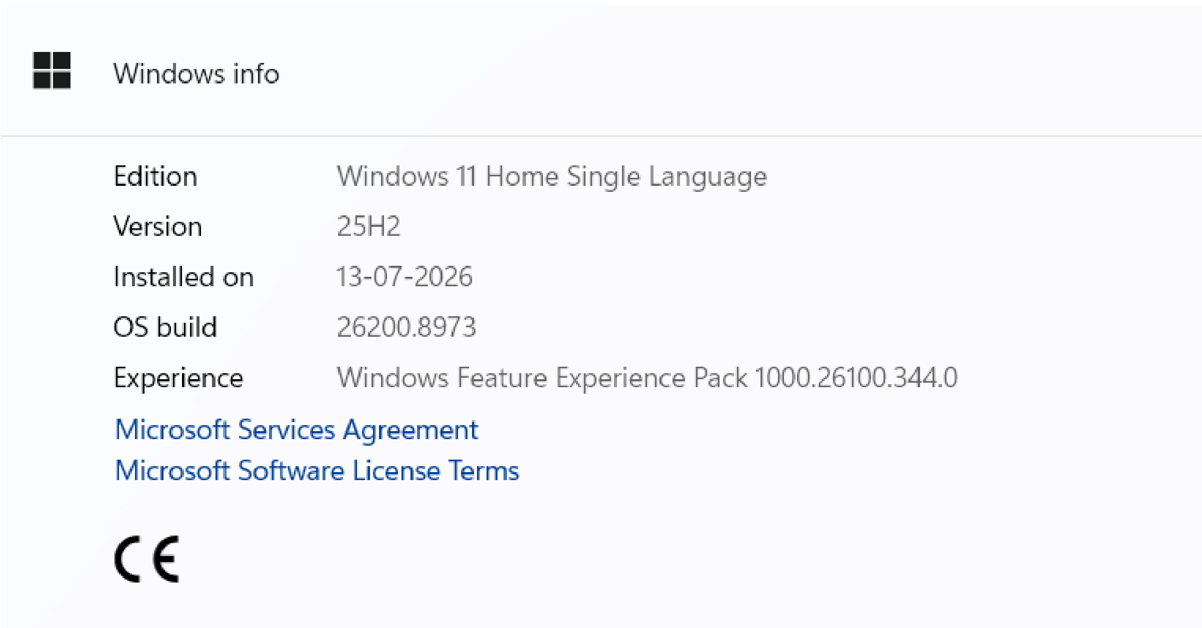
ASSIGNMENT 2.5

Q1. On your Windows laptop or PC, open 'System Information' and take a screenshot showing your OS version, processor, and RAM details.





Q2 **Download any official Linux distribution ISO (like Ubuntu or Fedora), and use Rufus or BalenaEtcher to create a bootable USB drive for installation. Hint: If you don't have a spare USB, use a virtual machine tool like VirtualBox to mount the ISO and start the installation process virtually.**

Option 1: Using Rufus to create a Bootable USB Drive

1. Download Linux ISO: Go to the official Ubuntu website **(ubuntu.com**) or Fedora website (**fedoraproject.org**) and download the latest Desktop ISO file.
2. Download Rufus: Visit **rufus.ie** and download the lightweight Rufus tool.
3. Connect USB Drive: Insert a spare USB flash drive (minimum 8GB) into the PC.
4. Configure Rufus:
   - Open Rufus and select your USB drive under Device.
   - Click SELECT next to Boot Selection and browse for the downloaded Linux ISO file.
   - Change Partition scheme to GPT and Target system to UEFI (non CSM) (recommended for modern PCs like Intel 10th Gen).
5. Create Bootable Drive: Click the START button, choose "Write in ISO Image mode" if prompted, click OK on the data deletion warning, and wait for the green bar to show READY.

Option 2: Using VirtualBox (Virtual Machine - If you don't have a spare USB)

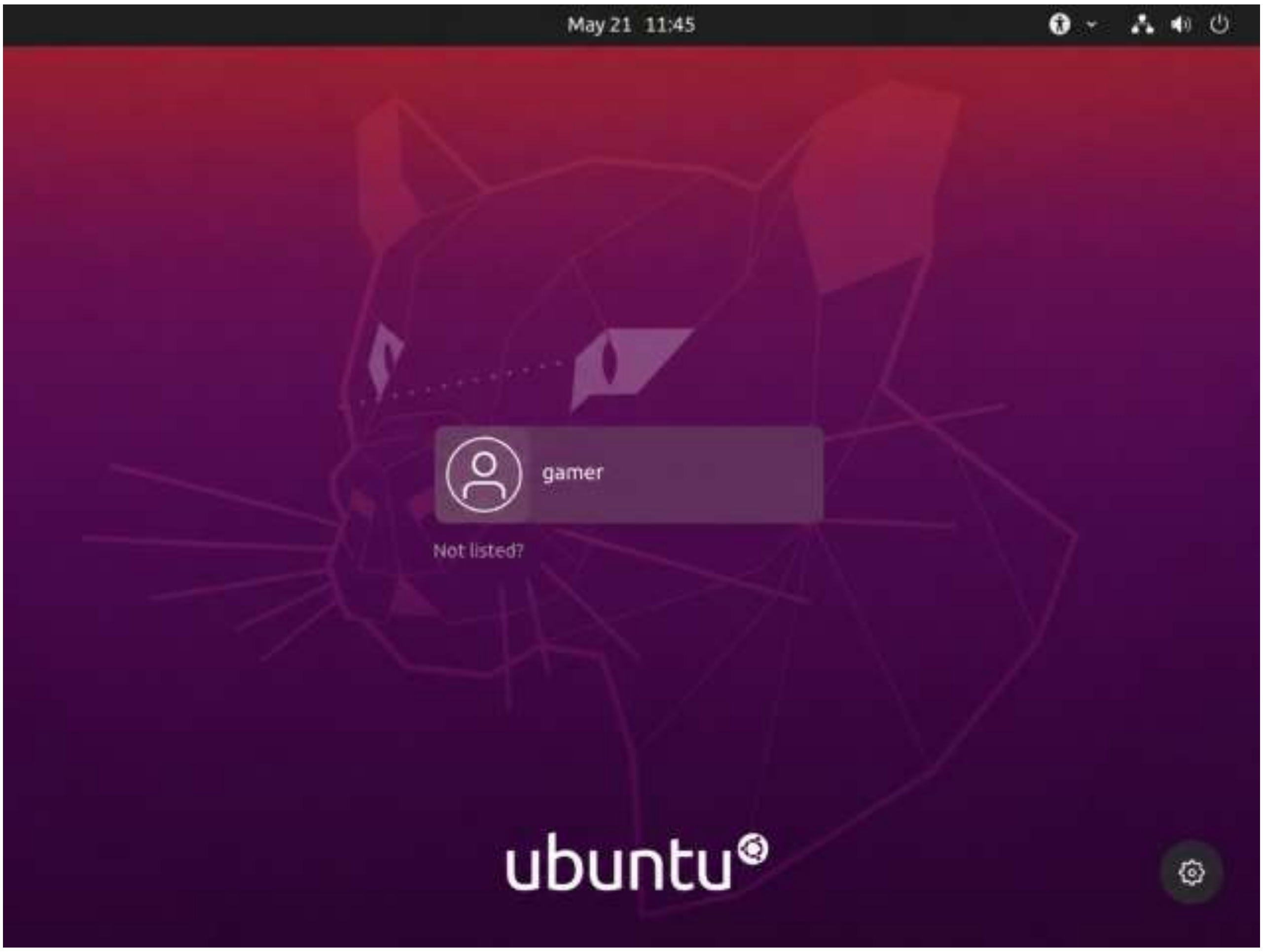
1. Download Tools: Download the official Linux ISO file and install Oracle VM VirtualBox from **virtualbox.org.**
2. Create New VM: Open VirtualBox, click New, give it a name (e.g., "Ubuntu"), and under the ISO Image option, select your downloaded Linux ISO file.
3. Allocate Hardware:
   - RAM: Allocate at least 2GB to 4GB of memory.
   - Processor: Assign at least 2 CPUs for smooth performance.
4. Create Virtual Disk: Choose "Create a Virtual Hard Disk Now" and allocate around 25GB of space (it will only use space inside the virtual system).
5. Start Installation: Select the newly created virtual machine from the left menu and click the green Start button. The Linux installation wizard will launch virtually inside Windows without affecting your main operating system.

Q3. **On a macOS device (or using a macOS virtual machine), open 'System Preferences' and change the desktop wallpaper, then adjust the system language to Hindi or Gujarati. Hint Find 'Language & Region' in System Preferences to add or change languages.**

Changing Wallpaper:

Changing Language:

Q4. Install a fresh copy of Ubuntu Linux (real or virtual machine) and create a new user account named 'gamer'. After installation, enable automatic login for this user and take a screenshot of the login screen showing the username.



Q5.Use ChatGPT to generate a step-by-step guide for installing Windows 10 on a laptop, including partitioning tips and post-installation settings for privacy and updates. Follow the guide and note any steps that were unclear or missing.

Step-by-Step Guide: Clean Installing Windows 10 on a Laptop

Important: Windows 10 is no longer generally supported. Before proceeding, check whether the laptop can run Windows 11. If it can, Windows 11 is the safer choice. If you specifically need Windows 10, consider enrolling in Microsoft's Extended Security Updates (ESU) after installation.

1. Back up everything

- ☐ Copy documents, photos, downloads, browser data, and other important files to an external drive or cloud storage.
- ☐ Export anything that cannot easily be recovered, such as application settings, saved game data, SSH keys, or locally stored passwords.
- ☐ Make sure you have installers/license information for software you will reinstall.
- ☐ If the laptop uses BitLocker/device encryption, make sure you have the recovery key.
- ☐ If possible, create recovery media from the laptop manufacturer.

Do not skip this step. A clean installation can remove personal files, applications, manufacturer customizations, and settings.

2. Check the Windows license

Before formatting the laptop, determine which edition you are licensed for:

- Windows 10 Home → install Home.
- Windows 10 Pro → install Pro.

Many modern laptops have their Windows product key embedded in firmware, so Windows Setup may automatically activate after installation. Microsoft notes that a product key might not need to be entered when a digital license or embedded key is available.

3. Create the Windows 10 installation USB

Use Microsoft's official Windows 10 installation-media tool rather than an unofficial ISO source.

- ☐ Use another working Windows PC if necessary.
- ☐ Connect a **blank USB flash drive with at least 8 GB**.

- ☐ Remember that creating the installation media erases the USB drive.
- ☐ Download Microsoft's Windows 10 Media Creation Tool.
- ☐ Run it and choose **Create installation media for another PC**.
- ☐ Select the appropriate language, edition, and 64-bit architecture.
- ☐ Choose **USB flash drive**.
- ☐ Let the tool download Windows and make the USB bootable.

Microsoft's current instructions confirm the 8-GB USB requirement and warn that its contents will be deleted.

4. Prepare the laptop to boot from USB

- ☐ Plug the Windows USB into the laptop.
- ☐ Shut down the laptop.
- ☐ Turn it on and open the boot menu. The key varies by manufacturer; common examples are **F12, F9, Esc, or F8**.
- ☐ Select the USB drive.
- ☐ If necessary, enter UEFI/BIOS and change the boot order.

Prefer **UEFI** boot rather than legacy/CSM mode on a modern laptop.

5. Start Windows Setup

When Windows Setup appears:

1. Select language, time/currency format, and keyboard.
2. Select **Install now**.
3. Enter your product key if requested.
4. If you have a digital license and are asked for a key, choose **I don't have a product key** when appropriate.
5. Select the Windows edition matching your license.
6. Accept the license terms.
7. Choose **Custom: Install Windows only (advanced)** for a completely clean installation.

6. Partition the drive carefully

This is the most dangerous part of a clean installation.

If you want the simplest setup

If the laptop has one SSD and you do **not** need a separate data partition:

- ☐ Identify the internal SSD by its size.
- ☐ Delete the existing partitions **only after confirming that you have backed up everything important**.
- ☐ You should eventually see **Unallocated Space**.
- ☐ Select the unallocated space and click **Next**.
- ☐ Let Windows Setup create the required EFI/System, MSR, recovery, and Windows partitions automatically.

Recommended: Don't manually create EFI, MSR, or recovery partitions unless you have a specific reason. Windows Setup can create the required partition structure for you.

If you want a separate data partition

For example, on a 1-TB SSD you might choose:

- ~200–250 GB for Windows and applications.
- The remaining space for a `D:` data partition.

During Setup:

1. Create the Windows partition from the unallocated space.
2. Leave sufficient space for Windows updates and applications.
3. Install Windows to that partition.
4. After Windows starts, use **Disk Management** to create/format the remaining space as a data partition.

Don't make the Windows partition unnecessarily small. A larger Windows partition gives applications, temporary files, updates, and recovery tools room to grow.

Important partition warning

Do **not** delete partitions blindly based only on their names. If the laptop has multiple physical drives, verify the drive number and capacity before changing anything.

If you see an OEM recovery partition and want to retain the manufacturer's factory-recovery option, don't delete it until you have confirmed that you have alternative recovery media.

7. Complete Windows Setup

After selecting the installation partition:

- ☐ Allow Setup to copy files and restart several times.
- ☐ Remove the USB when the laptop restarts if it keeps booting back into Setup.
- ☐ Select your region and keyboard.

- ☐ Connect to Wi-Fi/Ethernet when appropriate.
- ☐ Sign in with a Microsoft account or use the available local-account option for your edition/setup path.
- ☐ Review Microsoft's privacy and personalization choices rather than accepting every optional feature automatically.

8. Configure privacy

After reaching the desktop, open:

Start → Settings → Privacy

Review the categories individually.

Diagnostics & feedback

Go to:

Settings → Privacy → Diagnostics & feedback

Choose the minimum diagnostic-data option available to you. Microsoft confirms that Windows 10 exposes diagnostic-data controls here.

Advertising ID

Go to:

Settings → Privacy → General

Turn off:

Let apps use advertising ID to make ads more interesting to you based on your app activity

Microsoft notes that disabling this does not eliminate advertisements; it prevents apps from using Windows' advertising ID for this personalization mechanism.

Location

Go to:

Settings → Privacy → Location

- ☐ Turn off location access if you don't need it.
- ☐ Alternatively, leave location enabled but restrict access to apps that genuinely require it.

Camera and microphone

Go to:

Settings → Privacy → Camera

and

Settings → Privacy → Microphone

Review which applications have access and disable access for apps that don't need it.

Other privacy permissions

Also review:

- Account info
- Contacts
- Calendar
- Call history
- Notifications
- Documents
- Pictures
- Videos
- File system

Avoid disabling security-related functionality simply for the sake of privacy; review each permission according to how you use the laptop.

9. Install drivers

First install Windows updates that are actually available.

Then check the laptop manufacturer's support page for:

- Chipset drivers
- Graphics drivers
- Wi-Fi/Bluetooth drivers
- Audio drivers
- Touchpad drivers
- Storage/controller drivers
- BIOS/UEFI firmware updates

Avoid random "driver updater" applications. Prefer Windows Update and the laptop manufacturer's official support site.

10. Check Windows Update

Open:

Settings → Update & Security → Windows Update

Select **Check for updates** and install everything offered.

Restart and check again until there are no additional applicable updates.

However, this step has a major limitation in 2026: **Windows 10's normal support ended on October 14, 2025.** Microsoft says Windows 10 no longer receives normal free security fixes and other updates through Windows Update.

If you are intentionally remaining on Windows 10, investigate Microsoft's **Windows 10 Consumer Extended Security Updates (ESU)** program rather than assuming Windows Update will keep the machine secure.

11. Configure Windows Security

Open:

Settings → Update & Security → Windows Security

Check that:

- ☐ Microsoft Defender Antivirus is enabled.
- ☐ Firewall is enabled.
- ☐ Real-time protection is enabled.
- ☐ SmartScreen/security protections are enabled unless you have a specific reason to change them.
- ☐ Windows Security reports no immediate problems.

Don't install multiple real-time antivirus products unless you understand their interaction.

12. Finish the post-install setup

- ☐ Set a strong account password/PIN.
- ☐ Enable device encryption/BitLocker if supported and appropriate.
- ☐ Store the BitLocker recovery key somewhere safe.
- ☐ Set the correct time zone.
- ☐ Install your required applications from trusted sources.
- ☐ Restore your personal files from backup.
- ☐ Configure your browser's privacy settings.
- ☐ Create a fresh backup once the system is configured.

13. Verify the installation

Press **Windows + R**, type:

winver

and confirm the installed Windows version.

Also check:

Settings → System → About

and verify:

- Windows edition
- System type
- Installed RAM
- Device name

Open **Device Manager** and look for devices with warning symbols. Resolve those before considering the installation complete.

- **What was unclear or missing when I followed the guide**

1. The biggest missing piece: Windows 10 is now unsupported

Older Windows 10 installation guides commonly tell users to simply run Windows Update after installation. That advice is incomplete in 2026. Windows 10 Home/Pro 22H2 reached end of servicing on **October 14, 2025**.

A current guide must therefore explain Windows 11 compatibility and/or Windows 10 ESU.

2. Boot-menu instructions vary by laptop

There is no universal boot key. The exact key depends on the manufacturer and model. Microsoft's installation documentation similarly directs users to consult the PC manufacturer's instructions when the computer doesn't boot from installation media.

3. Partitioning needs stronger warnings

A generic "delete the old partitions and install Windows" instruction is potentially dangerous. A good guide needs to explain:

- Which physical disk is being modified.
- That deleting partitions destroys their contents.
- That EFI/MSR/recovery partitions normally should be created automatically.
- What to do if there are multiple drives.

- Whether the OEM recovery partition should be retained.
- How much space is sensible for a separate Windows partition.

4. Driver installation wasn't sufficiently emphasized

A fresh installation can leave Wi-Fi, graphics, audio, touchpad, or chipset devices without optimal drivers. The guide should explicitly include the manufacturer's support page and Device Manager verification.

5. Privacy settings vary by Windows version

The exact options and wording can differ between Windows releases and editions. Microsoft's current privacy documentation confirms the Windows 10 locations for Diagnostics & feedback and General privacy settings, but a guide should avoid promising that every listed toggle will appear identically on every installation.

6. "Install all updates" needs qualification

This was the most significant post-installation ambiguity. On a Windows 10 system in 2026, repeatedly checking Windows Update does **not** mean the system is receiving the same ongoing security support that a supported Windows release receives. ESU is the relevant exception for eligible users.